

Program: Diploma in Artificial Intelligence & Machine Learning	
Course Code: 3346	Course Title: Data Structure Lab
Semester: 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- Students will develop skills to efficiently implement linear and nonlinear data structures.

Course Prerequisites

Topic	Course name	Semester
Control Structures, Modular Program Design, Arrays, Functions	Problem solving and Programming	2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the concept of Pointers, Files and Structures in C	9	Applying
CO2	Make use of linear data structures to solve the problems	12	Applying
CO3	Implement searching and sorting algorithms	9	Applying
CO4	Make use of non linear data structures to solve the problems	12	Applying
	Lab Exam	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3	3	3	3	3	3	3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of the Experiment	Duration (Hours)	Cognitive Level
CO1	Demonstrate the concept of Pointers, Structures and files in C		
M1.01	Implement C programs to demonstrate Pointers.	2	Applying
M1.02	Develop C programs to demonstrate the concept of Structures.	2	Applying
M1.03	Develop C programs to demonstrate the concept of Files.	2	Applying
M1.04	Demonstrate Array based sequences & operations	3	Applying
CO2	Make use of linear data structures to solve the problems		
M2.01	Implement Stack using Array	3	Applying
M2.02	Implement Queue using Array	3	Applying
M2.03	Develop C program for Infix to Postfix conversion	3	Applying
M2.04	Implement Linked List operations - insertion, deletion, searching, etc. in C	3	Applying
	Lab Exam – I	1.5	
CO3	Implement searching and sorting algorithms		
M3.01	Implement Linear Search and Binary Search	3	Applying
M3.02	Implement Bubble Sort	2	Applying
M3.03	Implement Insertion sort	2	Applying
M3.04	Implement Quick sort	2	Applying

CO4	Make use of non linear data structures to solve the problems		
M4.01	Develop a program to create a Binary Search Tree	3	Applying
M4.02	Develop programs to perform the various traversals in a Binary Search Tree.	3	Applying
M4.03	Develop program to implement graph	3	Applying
M4.04	Implement the Depth first search Traversal / Breadth first search Traversal of graph	3	Applying
	Open Ended Experiments**	6	Applying
	Lab Exam – II	1.5	

**** - Suggested Open Ended Experiments**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open ended experiments as a group of 2-3. There is no duplication in experiments between groups.)

1. Implement traffic light software (Each light gets on one by one after a specific time interval.)
2. Shortest path between two nodes in a graph
3. Implement organizational chart using tree

Text / Reference

T/R	Book Title/Author
T1	Data Structures using C by E. Balaguruswamy - Tata McGraw-Hill Education
R1	Data Structures using C by Aaron M Tenenbaum - Pearson Education India
R2	Reema Thareja, "Data and File Structures using C", Oxford University Press.
R3	Jean-Paul Tremblay and Paul G. Sorenson, "An Introduction to Data Structures with Applications", Tata McGraw Hill.

Online Resources

Sl.No	Website Link
1	https://www.cet.edu.in/noticefiles/280_DS%20Complete.pdf
2	https://www.tutorialspoint.com/data_structures_algorithms/index.htm
3	https://www.sanfoundry.com/c-programming-examples-data-structures/
4	https://www.programiz.com/dsa